# CLEANING AND SUMMARIZING THE HUMAN INS GENE DATSET

## OBJECTIVE:

The objective of this task is to clean and summarize the human INS (insulin) gene dataset using basic bioinformatics analysis.

## DATA SOURCE:

- **Database:** NCBI Gene / NCBI Datasets
- **Gene Name**: INS (insulin)
- **Organism:** *Homo Sapiens*
- **Gene ID:** 3630
- **Dataset URL:** https://www.ncbi.nlm.nih.gov/gene/3630
- **File Format** : FASTA

## TOOLS USED:

- NCBI Gene / NCBI Datasets
- Python
- Google Colab

## METHODOLOGY:

The downloaded FASTA Dataset was analyzed using Python in Google Colab.

The following steps are performed:

- Loaded the FASTA dataset.
- Extracted individual DNA sequence records.
- Converted the sequences to uppercase.
- Calculated the sequence length.

- Counted the A, T, G and C nucleotides.
- Checked for ambiguous or other characters.
- Calculated the GC content percentage.
- Saved the Sequence statistics in a CSV file.
- Saved the cleaned sequences in FASTA format.

**RESULTS:**

Two FASTA sequence records were analyzed.

| PARAMETER | RECORD 1 | RECORD 2 |
|---|---|---|
| Sequence Length | 1,431 Bases | 1,431 Bases |
| A Count | 247 | 246 |
| T Count | 259 | 259 |
| G Count | 456 | 457 |
| C Count | 469 | 469 |
| Other/ambiguous Characters | 0 | 0 |

| | | |
|---|---|---|
| GC content | 64.64% | 64.71% |

**CLEANING SUMMARY:**

A total of zero characters were removed because both sequences contained only A, T, G and C.

**CONCLUSION:**

The Human INS gene dataset was successfully analyzed using Python. Basic sequence statistics, including nucleotide counts, sequence length and GC content, were calculated for two FASTA records. No ambiguous or other characters were detected in either record. The summary statistics and cleaned sequences were saved for further review.

**OUTPUT FILES:**

1. INS_gene_summary (1).csv **Contains the sequence statistics and cleaning information.**
2. INS_gene_cleaned.fasta **contains the cleaned DNA sequences.**